



GEOTOOLKIT DEMO: 3D RESERVOIR SEISMIC INTERSECTION

BROWSE DEMOS

Surface Property

- ☒ Show Surface
☐ Show Intersection

Reservoir Property

- ☒ Show Reservoir
☐ Show Intersection

PORTFOLIO

Reservoir Geologist & Static Modeler

Conrad Mako Gas Field · Petrel · 3D Seismic · AI Inversion



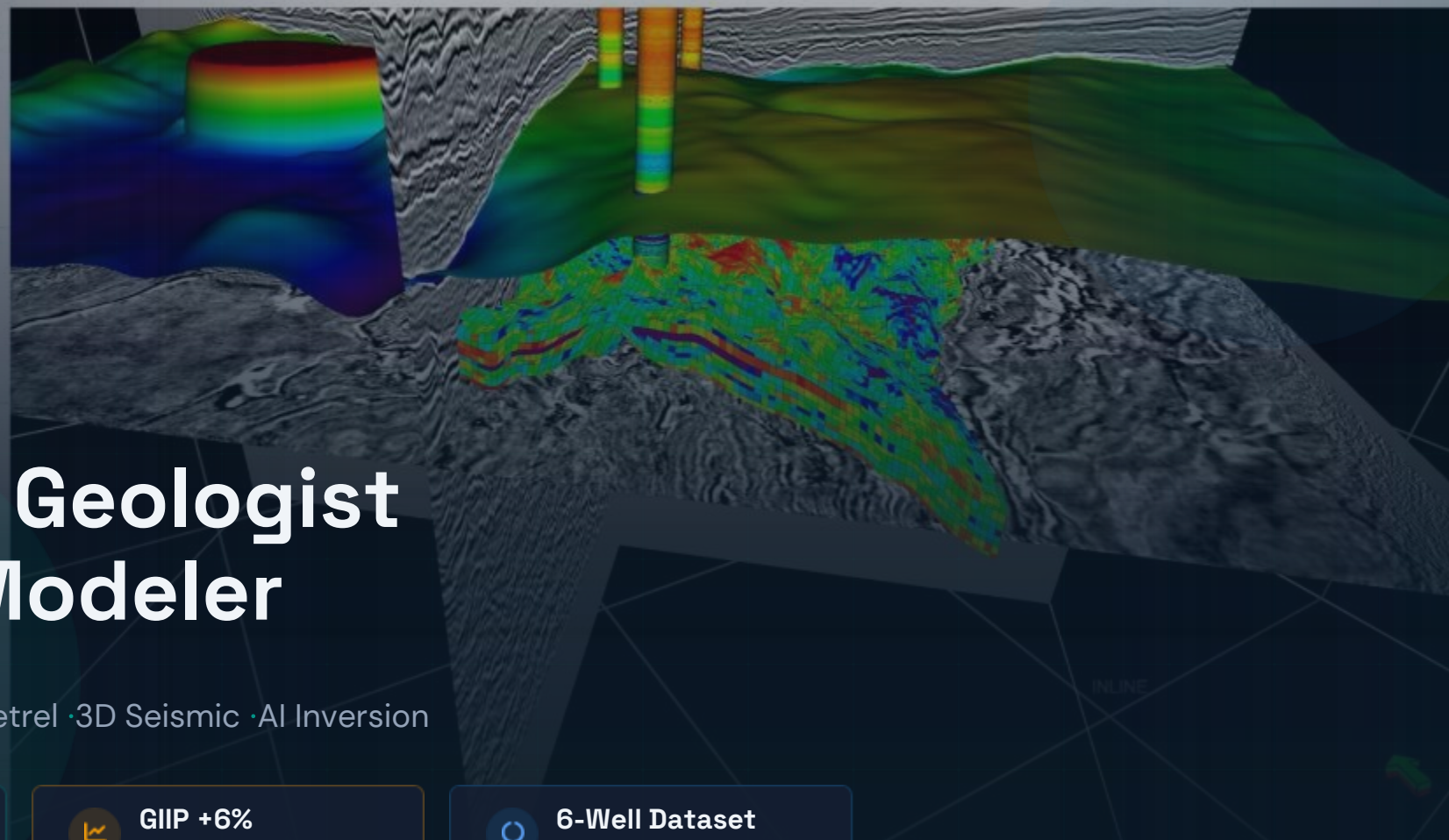
116 km² 3D Seismic
NEW DATA INTEGRATED



GIIP +6%
SIGNIFICANT GIIP UPLIFT



6-Well Dataset
MAKO FIELD FOUNDATION



Core Competencies



3D Seismic Interpretation

Interpreted reservoir tops and bases in Two-Way Time (TWT), then applied depth conversion across 116 km² of new 3D seismic data.

SEISMIC • TWT • DEPTH CONV.



Petrel Static Modeling

Executed full reservoir model update in Petrel following the Three60 standard workflow, integrating new structural and property data.

PETREL • THREE60 • STRUCTURAL



Facies & Property Modeling

Applied AI-guided Sequential Indicator Simulation (SIS) to model facies distribution and populate reservoir property grids.

SIS • AI-GUIDED • FACIES



Seismic AI Integration

Merged 3D Acoustic Impedance (AI) with legacy 2D Pseudo AI after rebalancing, ensuring seamless data continuity across the full field extent.

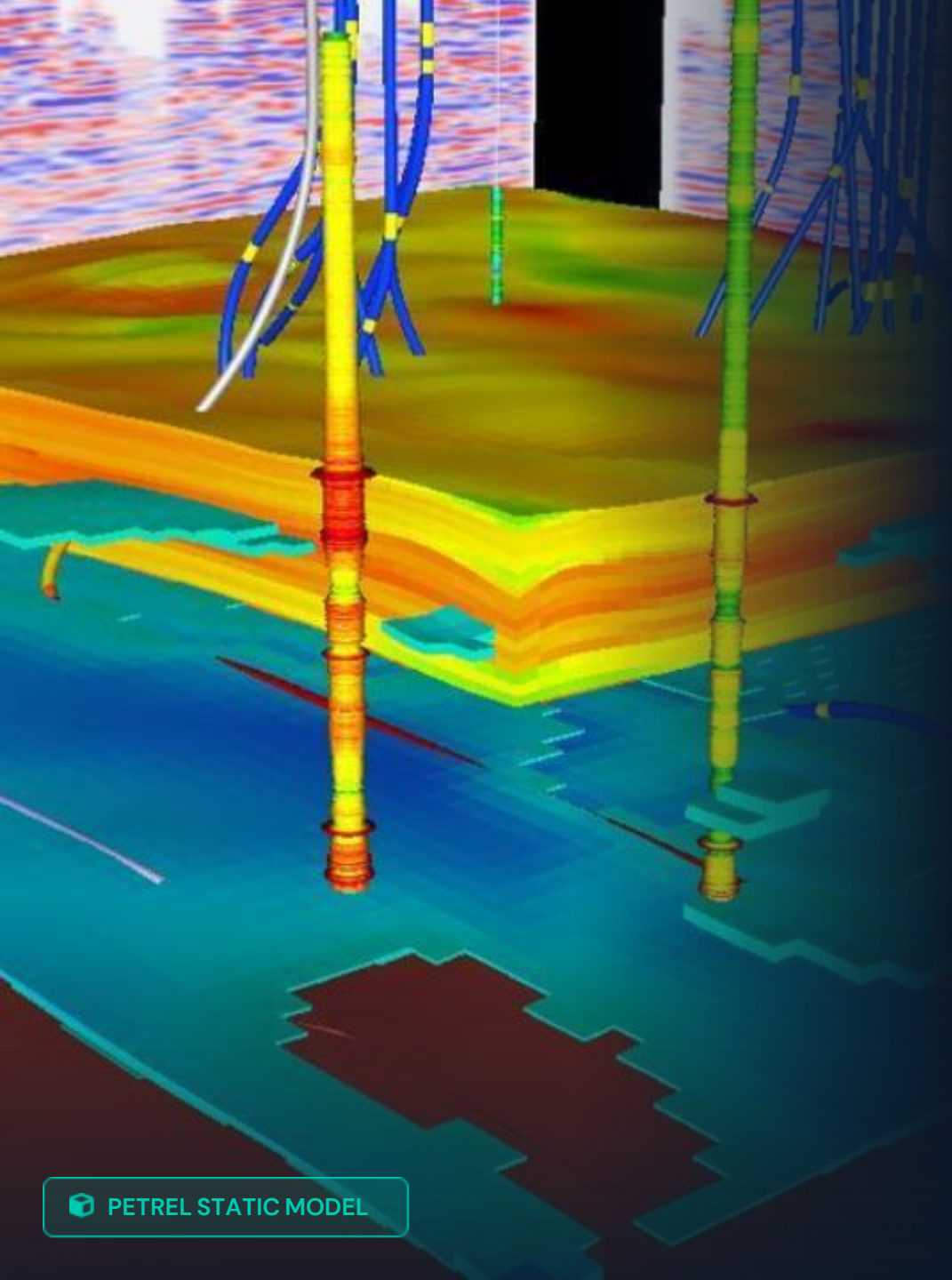
AI INVERSION • 2D/3D MERGE



GIIP Volumetric Estimation

Ran best-estimate volumetric workflow delivering a measurable uplift in gas-in-place, reflecting improved structural mapping and AI-guided facies quality.

GIIP • VOLUMETRICS • +6% UPLIFT



 PETREL STATIC MODEL

— FEATURED PROJECT

Conrad Mako Gas Field

Model Update 2023 — SW Area 3D Seismic Integration



3D SEISMIC COVERAGE

116 km² of new 3D seismic data integrated into the SW area with full inversion analysis



STRUCTURE UPDATE

Reservoir found up to 12m shallower than legacy 2D mapping— significant structural revision



2D / 3D DATA MERGE

3D AI merged with 2D Pseudo AI after rebalancing — seamless integration in Petrel reference project



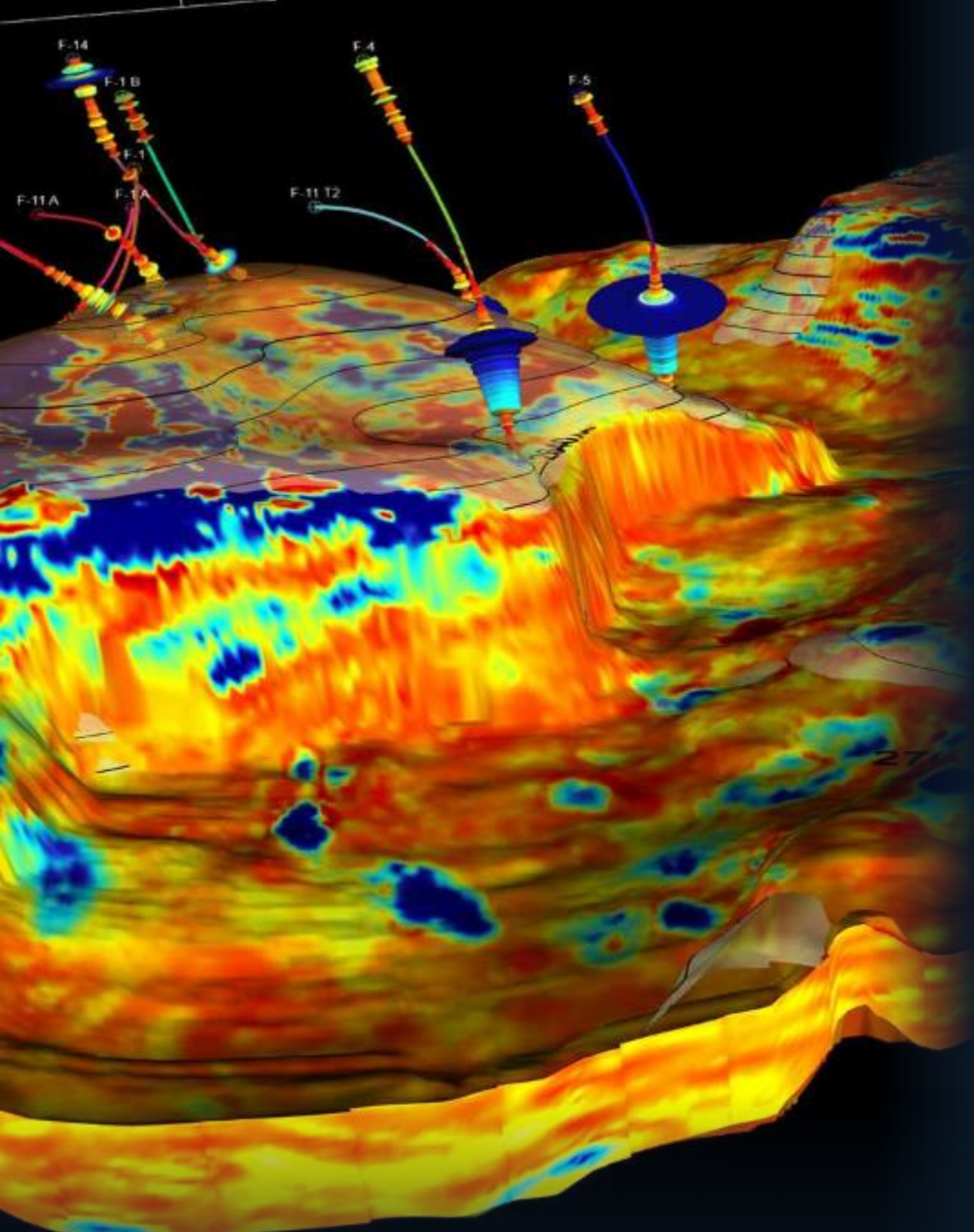
WELL DATASET FOUNDATION

6-well dataset underpins the static model — all wells originally drilled on 2D seismic interpretation

KEY OUTCOME — GIIP UPLIFT



Gas Initially In Place recorded a **+6% uplift following the updated structural model, confirming improved resource estimation across all reservoir zones**



● 3D SEISMIC VOLUME — AREA A (115.4 KM²)

SUBSURFACE MODELING

Seismic Interpretation & AI Integration



TWT Interpretation → Depth Conversion → Legacy 2D Merge

Reservoir tops and bases interpreted in 3D TWT domain, depth-converted, then merged with existing 2D legacy interpretations for full-field coverage.



AI Crossplot & Histogram QC

3D Acoustic Impedance rigorously compared with 2D AI using crossplots and histograms.

Area A: No rebalancing

Area B: Rebalanced



Rebalancing 3D AI to 2D AI Overlap

3D Pseudo AI (derived from 2D) merged with 3D AI after rebalancing. Moving average method applied for 2D AI population across the field.



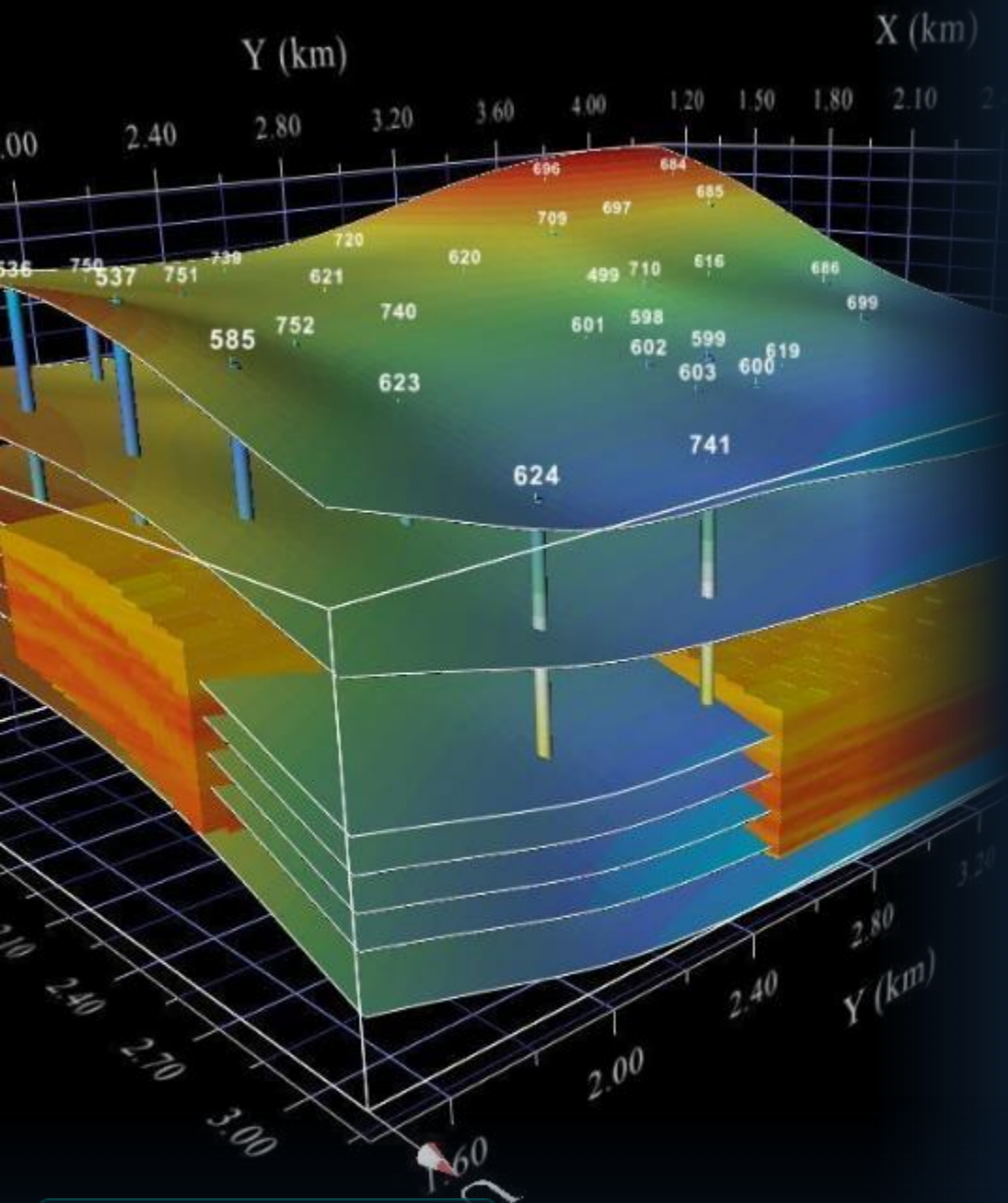
Anomaly Detection & Removal **DYG-13 Excluded**

DYG-13 identified as anomalous line and removed from the AI point set to ensure data consistency and prevent bias in facies population.

TECHNICAL PARAMETERS



3D Seismic Area: 115.405 km² (Area A) | **Method:** Moving average for 2D AI population | **Output:** Merged 3D + Pseudo AI dataset for facies guidance



3D FACIES MODEL — MAKO FIELD

STATIC MODEL UPDATE

Facies & Property Modeling

Three60 Standard Workflow · AI-Guided Simulation · Petrel Integration



AI-Guided SIS Simulation

Three60 workflow driven by seismic Acoustic Impedance — lower AI values in 3D area increased proportion of **good-quality reservoir facies** ; 2D area proportions remained stable.



Active Gas Zone Cells +4%

Updated model: **303,863 active gas cells** vs. legacy 291,739 — a 4% volumetric increase reflecting improved facies quality and structural update.



Porosity, Permeability & Sw Updated

Full property model suite (Porosity, Permeability, Water Saturation) rebuilt using standard workflow and integrated into the Petrel reference project.



Water-to-Gas AI Substitution

AI substitution applied in paleo-gas and water leg zones to correctly represent historical gas presence and improve property population accuracy.



Model Scale: Updated grid — 610,634 total cells vs. 585,501 legacy | **+4.3% cell count** driven by refined structural interpretation and AI-guided facies expansion.

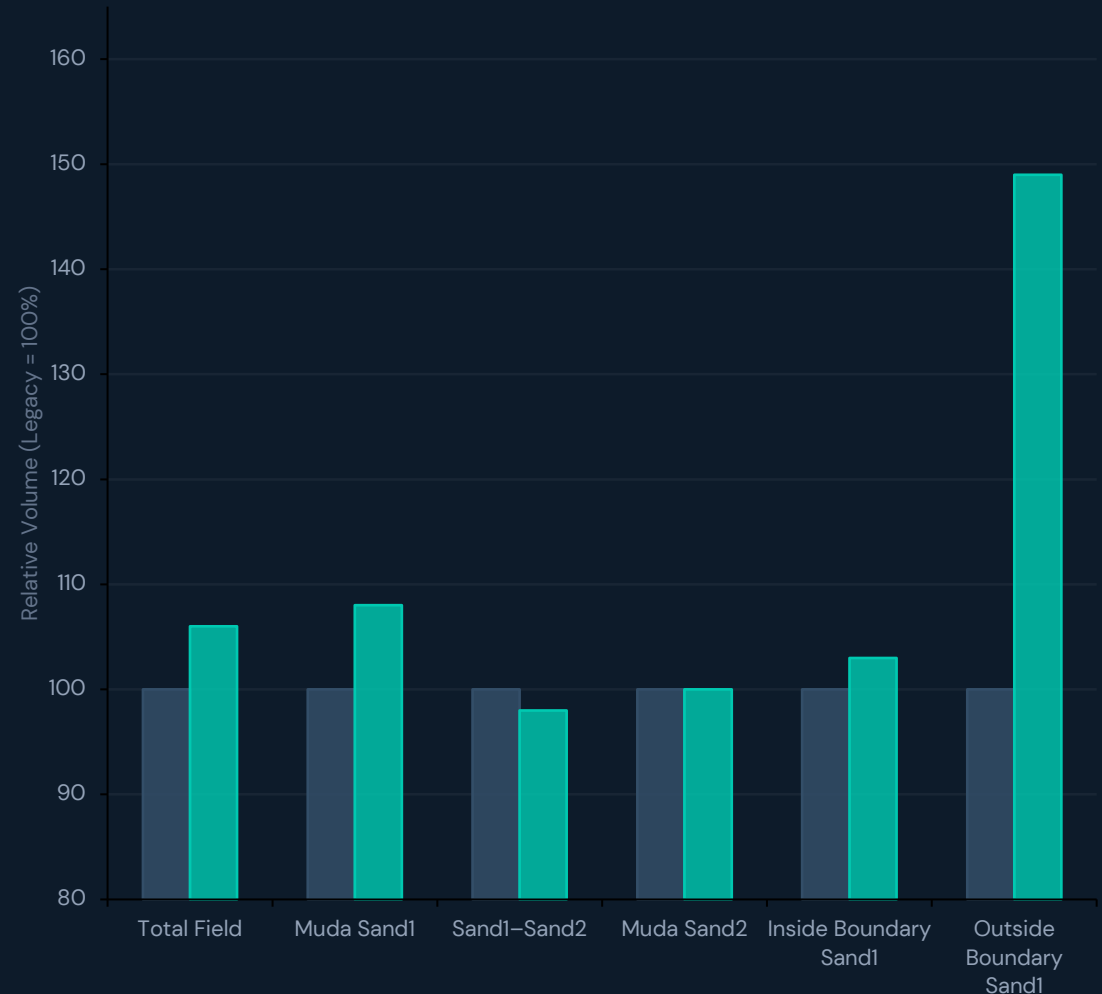
Volumetric Model Update

Static Model Update • Mako Field

Legacy Baseline (100%) Updated Model

GAS VOLUME CHANGE BY ZONE (INDEXED TO LEGACY = 100%)

Y-axis shows relative change vs. legacy model baseline



Key Findings

+6% Total Field — Field-wide gas volume increased across all zones, confirming the updated model delivers a stronger resource estimate than the legacy model.

+8% Primary Reservoir Zone — The main producing zone (Muda Sand1) recorded the largest absolute uplift, driven by the shallower updated structure.

+49% Upside Outside Boundary — The area outside the contract boundary showed the highest relative gain — nearly doubling the legacy estimate in that zone.

Uplift driven by **shallower structure (+12 m GRV)** and improved reservoir quality from AI-guided facies modeling.

KEY TAKEAWAY
The integrated 3D seismic and AI-guided modeling workflow delivered a measurable uplift across all reservoir zones — demonstrating the value of data-driven static model updates.

— PORTFOLIO • SUBSURFACE SPECIALIST

Let's Work Together

*Specialist in Static Reservoir Modeling & Seismic Interpretation
Available for consulting, contract, or full-time roles.*



Open to consulting & contract roles — flexible engagement across oil & gas projects worldwide.



Expertise in Petrel & subsurface modeling — 3D seismic interpretation, facies & property modeling, GIIP volumetrics.



Proven track record: 6% GIIP uplift delivered — measurable gas-in-place increase confirmed across a 116 km² 3D seismic, 6-well field model.

✉ GET IN TOUCH

Seismic AI Integration · Petrel Static Modeling
Subsurface Characterization · GIIP Volumetrics